

DECLAN KERR

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EDUCATION

University of Connecticut, Honors Program | 3.8 GPA

Storrs, CT

Bachelor of Science in Mechanical Engineering, Aerospace Concentration | Bachelor of Arts in German Studies

May 2029

- Eurotech Program: five-year dual-degree track including a six-month engineering internship in Germany
- **Honors:** Academic Excellence Scholarship, Dean's List (all semesters)
- **Relevant Coursework:** Fluid Dynamics, Thermodynamics I & II, Computer-Aided Design and Manufacturing, Vibrations, Circuits, Computational Fluid Dynamics, Materials Science & Eng., Computing for Mech. Eng.

WORK EXPERIENCE

Bauer, Inc.

Bristol, CT

Engineering Intern, Mechanical

Summer 2025, Winter 2025-'26, Summer 2026

- Supported design, build, testing, and checkout of hydraulic and pneumatic aerospace test stands by working directly with engineering and fabrication teams through the manufacturing lifecycle
- Performed component calibration and testing (e.g. pressure transducers, various valves, RTDs), analysis of test data, and assisted with plumbing, assembly, and troubleshooting on the shop floor
- Created and revised SolidWorks models, DraftSight drawings, and hydraulic and electrical schematic markups, along with test procedures and supporting engineering documentation

Kang Laser Research Group, University of Connecticut

Storrs, CT

Undergraduate Researcher, Advanced Laser Materials Processing and Energy Technology Lab

February - May 2026

- Developed Solidworks FEA simulations comparing shear performance of our novel laser-patterned Ti-6Al-4V/vitrimer adhesive joint to quantify strength enhancements from laser pattern
- Characterized Titanium substrate wettability via contact angle measurements to evaluate laser treatment effectiveness

PERSONAL TECHNICAL PROJECTS

8:1 Quasi-Direct-Drive Cycloidal Robotic Actuator

2026

- Designed a quasi-direct-drive robotic actuator from scratch, pairing a custom 8:1 cycloidal gearbox with a brushless motor and field-oriented controller across a 35-bearing drivetrain; iterated through three output architectures to a compact straddle-mounted design in printed carbon-fiber PETG
- Demonstrated full backdrivability and characterized the assembled actuator, measuring 3 N·m output torque and >50% transmission efficiency
- **Skills:** SolidWorks, Python (CadQuery parametric CAD, automated geometric verification), BLDC/FOC control, 3D printing, characterization/testing

Articulating Soldering Fume Hood

2026

- Designed and built a 3D-printed benchtop fume extractor for a personal soldering station, featuring a parallelogram-linkage arm on a c-clamp desk mount, fan-driven dual-filter box, and a magnetically detachable hood for filter swaps
- **Skills:** SolidWorks, kinematics, mechanism design, 3D printing and post-processing

Wireless Optical Flow Timer

2025-26

- Designed and fabricated a device to time the flow through a hose using a small light-based ESP32 circuit, which streams flow data over BLE to a custom SwiftUI iOS app
- **Skills:** Embedded C++, Swift, Bluetooth/BLE, circuit design and soldering, SolidWorks, fluid systems, 3D printing

SKILLS AND INTERESTS

Design & Analysis: SolidWorks (modeling, FEA), DraftSight, MATLAB, tolerancing, parametric CAD

Fabrication & Hardware: Hydraulic and pneumatic systems, shop-floor assembly, FDM 3D printing, soldering, Arduino/ESP32 circuits, motor control, BLE

Programming & Languages: Python, C++, Swift, Arduino IDE; German (intermediate)

Activities & Interests: UConn Boxing Team, Alpha Delta Phi fraternity; building, camping, reading, travel, running, modeling sports